Philosophy of Science
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Lecture 6
Paradigms and Revolutions

Motivation of this lecture

- Why do you need to know about paradigms and scientific revolutions?
- Because they are…
  - useful conceptual tools for making sense of scientific change
  - productive sources of metaphors in public discourse—e.g., “paradigm shift”
- Many writers use these concepts
  - You need to understand these references

Plan of this lecture

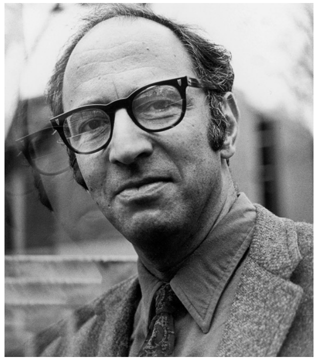
- Introduction to:
  - Thomas S. Kuhn
  - *The Structure of Scientific Revolutions*
- Key concepts
  - Theory ladenness of observation
  - Paradigm
  - Scientific revolution
  - Incommensurability

Reading for this lecture

- Bojana Mladenović, *Kuhn’s Legacy: Epistemology, Metaphilosophy, and Pragmatism*
  - Columbia University Press, 2017
  - Read chapter 1, “An Overview of Kuhn’s Philosophy of Science”, pp. 8–24
  - Available on Brightspace

Thomas S. Kuhn, 1922–1996

- Born in Cincinnati
- PhD in physics, Harvard University, 1949
- Developed a "General Education in Science" programme, Harvard, from 1950
- Professor of philosophy at MIT from 1979



Kuhn's principal work

- *The Structure of Scientific Revolutions*
  - 1st edition, 1962
  - 2nd, with "moderate" postscript, 1970
- Readable, short monograph
  - 173 pages, few footnotes and references
  - Sweeping, associative style
- Great scholarly and social influence
  - Most cited publication in humanities and social science between 1976 and 1983

Image of science before Kuhn

- Structure of science
  - Theoretical terms have clear and stable definitions. Examples: "atom", "gene"
  - Empirical data give univocal and objective verdicts about the adequacy of theories
- History of science
  - Growth of science is continuous and cumulative
  - All scientists in history share the same standards of rationality
- Example: Karl Popper's falsificationism

Image of science after Kuhn

- Structure of science
  - Definitions of theoretical terms change from one theory to another
  - Observations are partly shaped by theories: empirical tests are not independent of theory
- History of science
  - Development of science is discontinuous and non-cumulative: findings of one period do not hold in another
  - Standards of rationality change in time

Four key concepts

1. Theory ladenness of observation
2. Paradigm
3. Scientific revolution
4. Incommensurability
   - Semantic
   - Observational
   - Methodological

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Concept 1: theory ladenness

- We sometimes think of observational data as independent of theories
  - But this is naive
- All sense experiences are made and interpreted in the light of concepts and theories
  - We cannot “just look” independently of conceptual frames and theoretical assumptions

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Theory ladenness: example 1

- “I see a duck”
  - What could be simpler?
- But you cannot make this observation without having the concept of duck
  - Conceptual and taxonomic assumptions are involved in the simplest observation acts
- Observation is not purely perceptual…
  - … but both perceptual and conceptual

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Theory ladenness: example 2

- Observational acts in science also presuppose concepts
  - “The wind speed is 75 km/h”
  - “The elders conducted the initiation ritual”
- An additional form of theory ladenness is introduced by experimental apparatus
  - Observational acts then depend also on assumptions about the working of the apparatus

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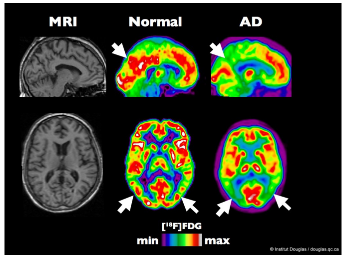
Theory ladenness: example 3



What we take a magnetic resonance imaging scan to show…

… depends on our assumptions about the workings of the scanner

Theory ladenness: implications

- "Pure observation" does not exist
  - All observations are contaminated with concepts and background theory
  - In part, rediscovery of themes from Immanuel Kant
- Theory test is trickier than we assumed
  - Jury is composed of observations + theories
  - Risk of circularity if observations are contaminated by the theory under test

Concept 2: paradigm

- We sometimes view scientists as formulating theories with an open mind
  - As in Popper's "context of discovery"
  - But this is naive
- Being a scientist involves signing up to a framework
  - Package of assumptions, techniques, and exemplars
  - Kuhn calls this "paradigm"

More about paradigms

- A paradigm…
  - is a conceptual framework that shapes scientists' thought and work
  - dominates a branch of science for a period
  - Kuhn calls such a period "normal science"
- A paradigm consists of:
  - Assumptions about the world under study
  - Exemplars for problem solving
  - Style of theorising

Paradigm: an example

- Newtonianism in physics, 1700–1900
  - Good example of paradigm in Kuhn's sense
  - Based on mechanics and theory of gravity of Isaac Newton (1642–1727)
- Components:
  - Metaphysics: the world consists of particles in motion under forces, such as gravity
  - Exemplars: how Newton made advances
  - Style of theorising: formulate mathematical laws to describe forces

Life within a paradigm

- Scientists solve "puzzles" by imitating exemplars
  - Assumptions of the discipline remain beyond dispute
- A paradigm brings…
  - Clear standards of progress
  - Professional stability and career structure
  - Coordination and concentration of effort
- In return, the paradigm imposes…
  - Strict bounds on creativity

How paradigms end

- Scientists encounter radically new data
  - These data cannot be explained within the constraints of the paradigm
- Successive stages: scientists…
  - First dismiss the new data as an anomaly
  - Then accept the need to explain the data
  - Finally criticise the established paradigm for its inability to account for the data
- This leads to a scientific revolution

Concept 3: scientific revolution

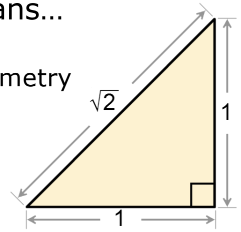
- Paradigm switch in a scientific discipline
  - Some scientists place the new data at centre of their thinking
  - They sketch a new paradigm on that basis
- Period of revolutionary crisis
  - Scientific community splits into conservative and progressive factions
  - Each can claim good reasons—paradigm switch is not fully rationally justifiable
- Revolution brings rise of new paradigm

Scientific revolution: example

- Transition from classical to quantum physics, 1925–1930
- Conservative faction
  - Albert Einstein, Erwin Schrödinger
  - Emphasised need to maintain determinism and visualisability of classical theories
- Progressive faction
  - Niels Bohr, Werner Heisenberg
  - Gave priority to maximising empirical performance with minimal metaphysics

Concept 4: incommensurability

- Kuhn writes that successive paradigms are incommensurable
  - … not just different or incompatible
- Incommensurability means…
  - Lack of common measure
  - Concept originated in geometry
  - No integers *m* and *n* such that $\sqrt{2} = m/n$
- Thus, revolutions are deep discontinuities



Forms of incommensurability

- Semantic
  - Pertaining to meaning of terms
- Observational
  - Pertaining to sense experiences
- Methodological
  - Pertaining to standards of rationality and progress

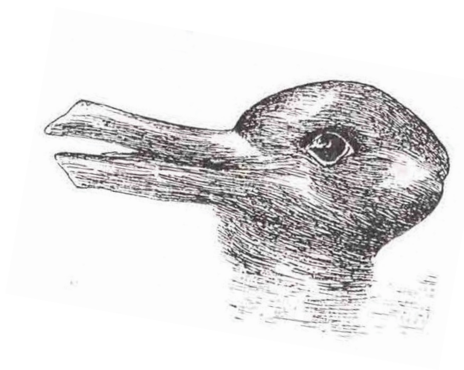
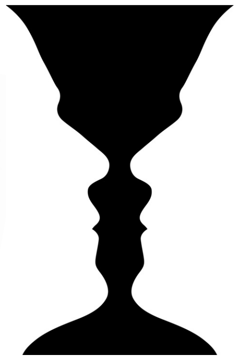
Semantic

- Scientific terms gain their meaning from a paradigm
- Example: “motion”
  - Aristotle: goal-directed qualitative change
  - Newton: quantitative change of position
- Consequences
  - Impossibility to translate among paradigms: communication failure
  - No clear logical relations between paradigms: not simply *P* and not-*P*

Observational

- Our paradigm determines what we see
  - Strong form of theory ladenness of observation
- In different paradigms…
  - we see different things
  - Or do we also live in different realities?
- Experience of *Gestalt* switch
  - Ambiguous figures: duck/rabbit (Joseph Jastrow, 1899), vase/faces

Gestalt switch: examples

Methodological

- Changes of reasoning styles
  - Example: reasoning in quantum physics
- Each paradigm has its own rationality
- No paradigm-independent…
  - criteria of theory choice
  - standards of scientific progress
- Unclear whether a later paradigm represents progress over an earlier
  - Is quantum physics better than Newtonian physics? Difficult to establish

Kuhn’s later view

- Partial retraction
  - Kuhn, *The Essential Tension*, 1977
- Five epistemic values have validity across paradigms
  - Accuracy, consistency, scope, simplicity, fruitfulness
- They assure continuity and progress of paradigm succession
  - However, application of these values depends on interpretation and arbitration

Kuhn’s legacy

- Awareness of ways in which observation depends on concepts and frameworks
  - … and rediscovery of Kantian themes
- Idea of plurality of scientific styles
- Problematising of concepts of scientific rationality and progress
- Increased attention for science’s…
  - Historical periodisation
  - Social dynamics

Recapitulation

- Thomas S. Kuhn
  - *The Structure of Scientific Revolutions*
- Theory ladenness of observation
- Paradigm
  - Frameworks strictly govern normal science
- Scientific revolution
  - Paradigm switch is not rationally justifiable
- Incommensurability
  - Breakdown of comparability

Picture credits

- Portrait of Thomas S. Kuhn: http://en.wikipedia.org/wiki/File:Thomas_Kuhn.jpg
- MRI/PET scan: http://www.photoree.com/photos/permalink/1654898-8426710@N02
- Photograph of Philips MRI scanner: http://commons.wikimedia.org/wiki/File:MRI-Philips.jpg
- Square root of 2:
- http://commons.wikimedia.org/wiki/File:Square_root_of_2_triangle.png
- Duck–rabbit:
- http://mysituationalawareness.com/tag/bus
- Vase/faces: http://www.crafthubs.com/the-white-vase-trick/26883